Moral Tensors and DecisionProofs: Compiling Language into an Auditable, Grounded Safety Layer

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Abstract

Most AI safety is a scalar at the output: one good/bad check before an action ships. We keep the structure instead — who is affected, what is owed, who consented, who bears imposed risk — and contract to a decision only at the end, auditably. Two open-source pieces realize this. `erisml-compiler` turns natural-language moral material into a typed MoralGraph and a rank-1…6 moral tensor, read through four ethical lenses at once; when the lenses disagree it refuses to silently aggregate and defers to a human. At runtime, ErisML's three-layer gateway drops the same evaluation into an agent's plan→act loop, sealing each decision in a SHA-256 hash-chained DecisionProof and failing safe, never open. The geometry grounds it: correlative swap and deontic negation are commuting involutions on Hohfeld's normative positions, generating the *measured* Klein four-group V_4; the full dihedral D_4 is a posited, testable extension, not an established result — a correction we made to our own earlier claim and document here. A Bond Index tests whether a judgment survives the agent↔patient swap. The PyTorch hook is literal: forward hooks on transformer layers compare what a model *says* with what it internally *exhibits*. Running today on a Qwen/Gemma agent stack and two governed AI NPCs.

1 Structure before the scalar

A single good/bad score discards exactly the information an audit needs: which stakeholders are affected, what is owed to each, who consented, and who bears imposed risk. ErisML keeps that structure as a typed object through the whole evaluation and performs *decision contraction* only at the end — logging the weights used, who lost, and what moral residue remains. The claim is not that the system knows what is good; it is that every collapse from structure to verdict is recorded and replayable.

2 The compiler

`erisml-compiler` (alpha v0.9.0, PyPI, MIT, 477 tests, Zenodo DOI 10.5281/zenodo.20659432) runs a 12-pass pipeline: `text` → `segment` → `extract` → `canonicalize` → `MoralGraph` → `tensorize` → `DEME` → `DecisionProof`. The MoralGraph is typed (stakeholder, act, maxim, commitment, fact, norm nodes) and carries a canonical SHA-256 hash. Extraction has three tiers: deterministic rules, an LLM tier (vLLM with a critic pass), and a LaBSE classifier-head probe.

Framework pluralism. One graph is read through four projections — consequentialist (harm tensor; Gini, worst-off, exact Shapley attribution), deontic (Kantian universalizability gates formulated in Z3 SMT: an UNSAT core witnesses a contradiction in conception), virtue, and care. The projections are not averaged. When they disagree, `cross_projection_disagreement` fires and the decision is deferred to a human. We treat this as the responsible default: the compiler does not pick a moral theory and hide the choice.

3 The runtime

`erisml-lib` (v3.0.0, source install) provides the three-layer runtime pipeline between an agent's planner and its actuators: *Reflex* (fast hard stops), *Tactical* (full MoralVector reasoning, 10–100 ms), *Strategic* (policy recording and human oversight), emitting `ALLOW` / `REVISE` / `BLOCK`. Every decision is sealed in a DecisionProof whose SHA-256 `proof_hash` chains to the previous proof and to the IR hash of the compiled material, so a judgment can be replayed and verified. If the ethics service is unavailable, the gateway falls back to rule-based checks — it never fails open. (The Z3 gate is compiler-side; the runtime's deontic gate is a dependency-free rule table.)

4 The geometry, corrected

Hohfeld's first-order normative positions — Obligation, Claim, Liberty, No-claim — sit on a square with two semantically motivated operations: the *correlative swap* s (agent↔patient: O↔C, L↔N) and *deontic negation* r^2 (O↔L, C↔N). These are *commuting involutions*, so the group they generate is the Klein four-group $V_4 = \{e, s, r^2, sr^2\}$ — abelian, order 4. That is the **measured** structure.

Our earlier materials (including the originally submitted version of this poster) claimed the full dihedral group D_4 (order 8, non-abelian). That was an overclaim, and we corrected it in July 2026: D_4 is **posited**, licensed only if quarter-turn operations (r, r^3, sr, sr^3) are independently demonstrated as normative operations, which has not been done empirically. The library implements the full D_4 machinery so the hypothesis stays testable, with a regression test asserting that the implemented operations generate exactly V_4, and a machine-checked Lean 4/Mathlib proof (`formal/HohfeldV4.lean`) verifying the commuting involutions, the 4-element closure $\cong V_4$, and that the quarter-turn lies outside it. This is the second self-correction in the program's history: an earlier $SU(2) \times U(1)$ gauge hypothesis was falsified by CHSH-style tests on human moral judgments (N=600, all $|S| \leq 2$). We consider the corrections a feature of the method — the framework makes claims sharp enough to be wrong.

The Bond Index. B_d measures the correlative-symmetry defect: the fraction of scenario pairs whose verdicts fail to commute with the swap s. $B_d = 0$ is perfect agent/patient symmetry; the empirical human baseline is $B_d = 0.155$, estimated from a 20,030-letter advice-column corpus (1985–2017); the runtime warns at 0.25 and blocks at 0.30. The Bond Index depends only on the generator s — the one operation that is fully confirmed — so it is unaffected by the $D_4 \to V_4$ correction.

The same swap-based audit logic, generalized to a matched-neighbourhood disparity diagnostic (the Legal Bond Index), detects excess cross-race score divergence in the COMPAS data: LBI = 1.049 [1.027, 1.072] at k=20 (binary high-risk cut: 1.092), with a conditional randomization test rejecting at $z = 4.9$ (global multiscale $p = 0.001$), attenuating to ≈1.00–1.02 under matching-parity specifications (Zenodo DOI 10.5281/zenodo.21310251). It is offered as a research-stage screening signal, not a validated regulatory audit standard.

5 The PyTorch lens

The PyTorch hook is literal. Forward hooks on selected transformer layers feed an *activation lens* (what the model exhibits internally), compared against a *text lens* (what it says); a *delta lens* flags divergence and raises `requires_human_review` across five named failure modes. The adapters run against Hugging Face causal LMs, with Qwen- and Gemma-family models exercised in deployment. Caveat: the activation probe is *uncalibrated by default* — its numeric output is research-grade, a promising extension rather than a finished detector.

6 Worked example

The bundled `nazi_attic` case (sheltering refugees, asked a direct question) compiles to per-party verdicts: speaker harm 0.76 (*forbid*), village 0.83 (*forbid*), refugees 0.00 (*prefer*), nazis 0.18 (neutral); Gini(harm) = 0.43; worst-off = village; exact Shapley attribution per stakeholder; the DecisionProof's `proof_hash` chains to `audit.ir_hash`. One command, real numbers, a hash you can verify.

7 Deployment

The layer gates a live multi-agent stack (Qwen3/Gemma models on dual-GPU hardware) and two governed AI NPCs (ARTEMIS and SIGMA-4 in the Halyard RPG), each behind the validator, DecisionProofs, and a human keeper kill switch. Embodied robotics is the design target; conversational agents are the live deployment.

8 Epistemic status and limits

Alpha software (compiler v0.9.0); the rule/LLM text path is solid and tested; the activation lens is early and uncalibrated; tensor ranks 1–3 are fully real, higher ranks partial (the action axis is a stub); V_4 is measured, D_4 posited; the Hohfeld/V_4 module and Bond Index live in `erisml-lib`, not yet in the compiler. None of this is a solved-ethics claim — it is an auditability claim.

Availability

`pip install erisml-compiler`. Source: `github.com/ahb-sjsu` (`erisml-compiler`, `erisml-lib`, `agi-hpc`). Zenodo DOIs: 10.5281/zenodo.20659432 (compiler); 10.5281/zenodo.21310251 (COMPAS/LBI analysis archive).

References

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